

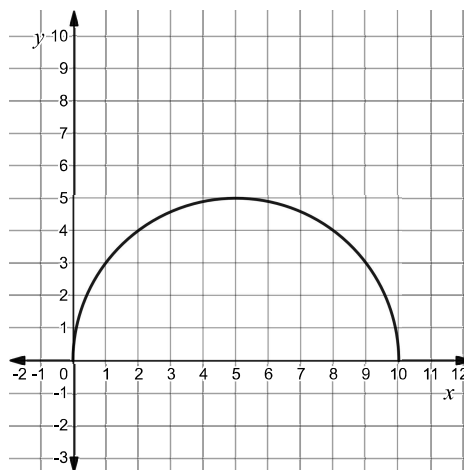
Precalc Lesson 1.2 Homework

Name _____

1. The complete graph of a function is shown below.

a. What is the domain of the function?

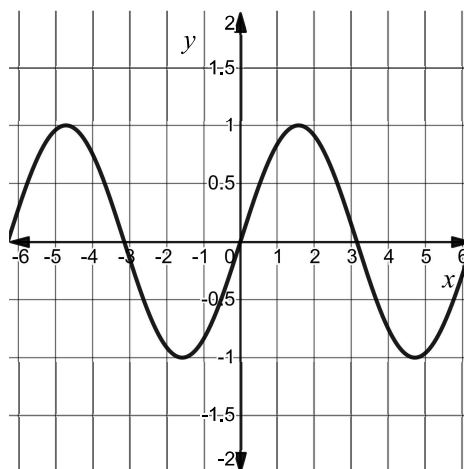
b. What is the range of the function?



2. A portion of the graph of $f(x) = \sin(x)$ is shown.

a. What is the domain of the function?

b. What is the range of the function?



3. Let $y = \frac{3}{2x - 6}$.

a. What is the domain of the function?

b. What is the range of the function?

4. Let $y = \sqrt{81 - x^2}$.

a. What is the domain of the function?

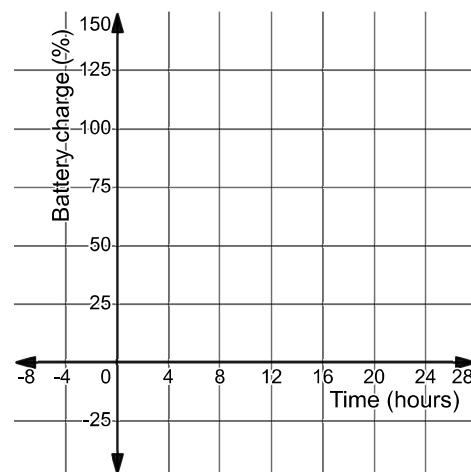
b. What is the range of the function?

5. $y = (x - 4)(x + 3)(x + 5)$

a. What is the domain of the function?

b. What is the range of the function?

6. The battery charge of a cell phone is tracked with a function over the course of one day. Graph a possible function below that correctly represents the domain and range of the situation.



7. Find the domain of $y = \frac{2}{(x - 3)(x + 7)}$.

8. The temperature of a mug of hot chocolate in degrees Fahrenheit is given by $H(t) = 70 + 100(0.819)^t$ where t is the time, in minutes, after it is prepared.

a. What is an appropriate domain for this situation? Explain your reasoning.

b. What is an appropriate range for this situation? Explain your reasoning.

9. A wedding caterer charges a \$200 service fee and \$24 per plate (cost of food for one guest). Johanna has gotten RSVPs from 72 guests but is still waiting to hear back from the other 18. She also has to pay for her own food and the groom's food, who are not considered guests. Let $C(p)$ represent Johanna's catering cost for p plates of food.

a. What is the domain of $C(p)$?

b. What is the range of $C(p)$?

10. Which of the following functions has a domain given by $\{x \in \mathbb{R} \mid x \neq 4\}$?

A) $f(x) = \sqrt{x-4}$

B) $g(x) = x^2 - 16$

C) $h(x) = \sqrt{9-x^2}$

D) $j(x) = \frac{2}{3x-12}$